DECISION 546Q
Section C Team 45
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DECISION 546Q: Final Project

A deep learning solution to predict song popularity for music streaming platforms

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In the modern era of music, online streaming platforms have become the central way to listen to music and explore new musical artists. Many of the big-name firms dominating this industry (Spotify, Apple Music, SoundCloud, etc.) constantly seek a competitive edge to improve revenue generation and user retention over their rivals. As a core element of the music industry, gaining insight into what music will be popular is essential to preventing user churn and maintaining market share. While there is no way to predict future song popularity with 100% certainty, deep learning can be utilized to provide reasonable estimates on the subject.

Model and Model Development:

In developing the most appropriate model to predict song popularity, it was important to consider several key features of the data set and impact of each individual musical component (feature) on the popularity value. For reference, the dataset "Song Popularity" contains approximately 15,000 entries and includes the following information for each entry: song name, song popularity (1-100), song duration, acousticness, danceability, energy, instrumentalness, key, liveness, loudness, audio mode, lyric density, tempo, time signature, and audio valence. In looking at the distribution, prior to the train and test split, it became apparent that the data set had a severe class imbalance. More specifically, the distribution showed far more high-popularity songs than low popularity songs. As a result, we decided to apply stratification to the test/train split, splitting the data into three bins (low 0-30, medium 30-60, and high 60-100). Without stratification, the model would see far too few low-popularity samples to accurately predict in that range. In addition, we tried applying a weighted loss function to the bins to manage the skewage when training, however, found that it had the opposite effect, actually decreasing overall model accuracy.

Another key consideration was the network structure itself. Initially, we started with a deep structure (512 → 256 → 128 → 64 → 32 →1), but found that with only 13 input features, a deeper network was overfitting. As a result, we moved to a shallower network with wider layers.

Our final model is a multilayer perceptron (MLP) neural network with a wider, shallower architecture, and approximately 137,000 trainable parameters. The architecture consists of three hidden layers, with batch normalization, ReLU activation functions, and dropout layers applied to enhance generalization and mitigate overfitting. The final output layer produces a single continuous value representing the predicted song popularity score on a scale of 0–100.

Layer Details:

1. Input Layer: 13 features derived from audio attributes such as energy, loudness, tempo, and valence.
2. Hidden Layer 1: 384 neurons → Batch Normalization → ReLU → Dropout (p=0.15)
3. Hidden Layer 2: 192 neurons → Batch Normalization → ReLU → Dropout (p=0.15)

4. Hidden Layer 3: 96 neurons → Batch Normalization → ReLU → Dropout (p=0.10)
5. Output Layer: 1 neuron (linear activation as a single popularity score)

We used Mean Squared Error (MSE) as the primary loss function, which aligns with our regression objective of minimizing the squared difference between predicted and true popularity scores.

Several intentional design choices were made to enhance model performance and generalization. First, batch normalization was added after each linear layer to stabilize training and improve convergence. Second, dropout layers were incorporated with decreasing rates (0.15, 0.15, 0.10) across successive hidden layers to reduce overfitting. Third, the architecture progressively reduces neuron counts (384→192→96) to gradually compress feature representations while maintaining predictive capacity. Finally, a cosine annealing scheduler with warm restarts was used to dynamically adjust the learning rate, helping the model avoid local minimum(s) during training.

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============================================================
MODEL PERFORMANCE SUMMARY
============================================================
Accuracy (±10 points): 46.32%
Mean Absolute Error:   14.790 popularity points
Root Mean Squared Error: 19.690 popularity points
R² Score:              0.220
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```

While the model performs significantly better than random guessing, the moderate R^2 value indicates that predicting song popularity from audio features alone remains challenging, as noted in our initial progress report. The architecture and training regimen were designed to balance model capacity with regularization, reflecting a systematic approach to a complex regression task.

Analysis and Observations:

Looking at the results of our model and data analysis, several interesting points stand out. First, even with several efforts to account for popularity skew, our final model's weakest area of prediction is in the low-popularity category. As indicated in the visualizations below, our model performed most accurately in the medium and high popularity ranges (Figs. 1 and 2). If the data set were larger and contained more low-popularity songs, our final model would achieve much higher accuracy. With this being said, given our model's high performance in the medium/high popularity subgroups, predicting the potential popularity of a song (yet to be released) of an established artist (Taylor Swift) would likely be highly accurate.

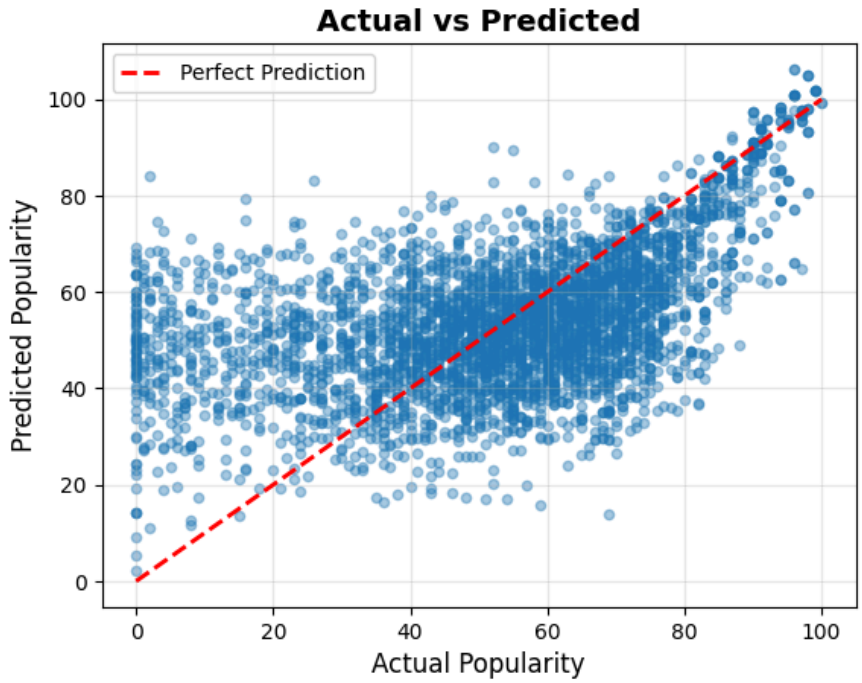

Fig 1. The visualization above shows our predicted values (blue) with respect to the actual or perfect prediction. As one can see, the model becomes more accurate when the popularity score is above 40.

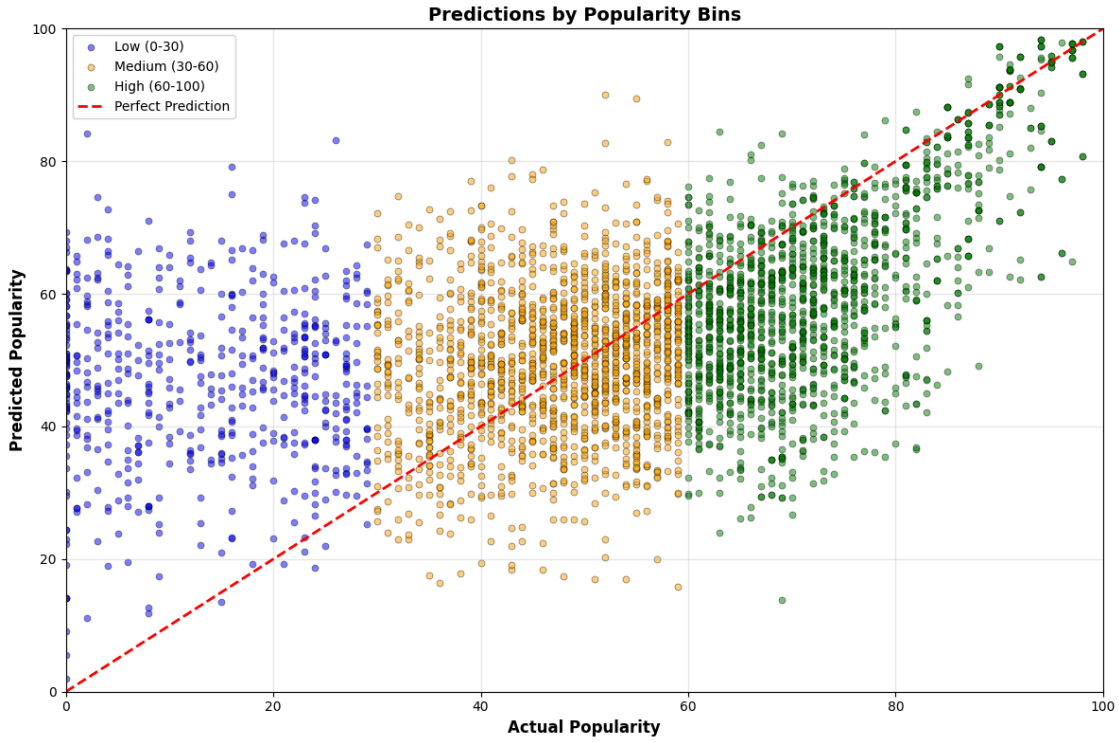

Fig 2. Similar to figure 1, the visualization above shows our predicted values (broken down by popularity bin) with respect to the actual or perfect prediction. As one can see our model is least accurate for scores in the low bin (blue).

Performance Comparison with Benchmark Model:

When comparing our deep learning model to the regression models used in the benchmark project (Appendix-1), the improvement in predictive accuracy becomes immediately clear. The classical approaches including Multiple Linear Regression, Ridge Regression, Lasso Regression, and Elastic-Net Regression all produced Test-RMSE values between 20.35 and 20.40, with extremely low R^2 scores (near 0.00) as shown in the table below. This indicates that the regression models explained almost none of the variance in song popularity.

In contrast, our deep learning model achieved a lower RMSE of 19.911 and a significantly higher R^2 of 0.202, meaning it captured over three times the explained variance of any regression model. Additionally, our model demonstrated greater error robustness through improved MSE and MAE values, and it achieved 44.89% accuracy within ±10 popularity points, which none of the regression models reported.

Polynomial Regression performed especially poorly, with RMSE exploding to 6.67×10^{10}, demonstrating catastrophic overfitting and confirming that linear models cannot capture the nonlinear complexity of the underlying feature space.

Overall, our deep learning architecture far outperforms all baseline regression models in terms of error minimization, variance explanation, and predictive strength, validating its suitability for predicting real-world song popularity on streaming platforms.

Baseline Model			
Model	**Test-RMSE**	**Test-MSE**	**Test-R^2**
Multiple Linear Regression (MLR)	**20.40289**	416.2779	≈0.003
Ridge Regression (RLR)	**20.40257**	416.2648	≈0.003
Lasso Linear Regression (LLR)	**20.36469**	414.7208	≈0.007
Elastic-Net Regression (ENR)	**20.35949**	414.5089	≈0.0076
Polynomial Regression (PNR)	**6.67×10^{10}**	4.45×10^{21}	$-1.06e^{8}$ (catastrophic overfitting)
Our Model	**19.911**	**396.457**	**0.220**

Business Impact and Applications: Marketing Cost Savings and ROI

- **Advantage Opportunity:** Platforms spend hundreds of millions annually promoting new music releases. Historically, marketing budgets are spread across all releases equally, often leading to wasted spending on songs that won't be popular.
- **Solution:** Our song popularity model can be used as a tool to identify high-potential songs before they're released, simply based on musical features. Deployed correctly, streaming platforms could reduce wasted promotional spending on songs unlikely to gain audience engagement.
- **Impact:** Shifting marketing expenditure based on popularity prediction could dramatically improve

To analyze the potential financial impact our model could have on the marketing budget of a large streaming platform, we calculated baseline performance/budget allocation and expenditures. We compared them with an optimized scenario in which the budget is allocated based on the identification of "high-potential" songs.

More specifically, we assumed that a hypothetical large streaming platform (operating similarly to Apple Music or Spotify) operates under a ~$500M marketing budget, spread equally across 12 million song releases (industry averages). More specifically, this baseline (without predictive modeling) allocates the annual budget evenly across every song, with an industry-average ROI of 1.5x ($1.50 for every $1.00 spent), generating ~$750M in annual revenue.

If our model were deployed at a company with similar constraints, ROI and Revenue could be improved significantly. More specifically, even with our model's accuracy, it could identify high-potential songs (high popularity scores) and allocate 70% of the promotional budget accordingly, rather than an even distribution across all songs/artists.

Based on our calculations, the deployment of our model has the potential to improve marketing/promotional ROI from 1.5x to 2.06x (per dollar spent) and to project an additional $280M in revenue. The details of our calculations and implementation are available below and in the attached Python project file.

```
Revenue from High-Potential Songs:  $805,000,000
Revenue from Remaining Songs:       $225,000,000
Total Revenue Generated:            $1,030,000,000

Previous Revenue (No Model):        $750,000,000
Additional Revenue (With Model):    $280,000,000
ROI Improvement:                    1.2%

New Marketing ROI:       $2.06 per $1 spent
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Fig 4. The excerpt above, directly pulled from our Python calculation results, shows the monetary difference between the current strategy and the proposed model deployment.

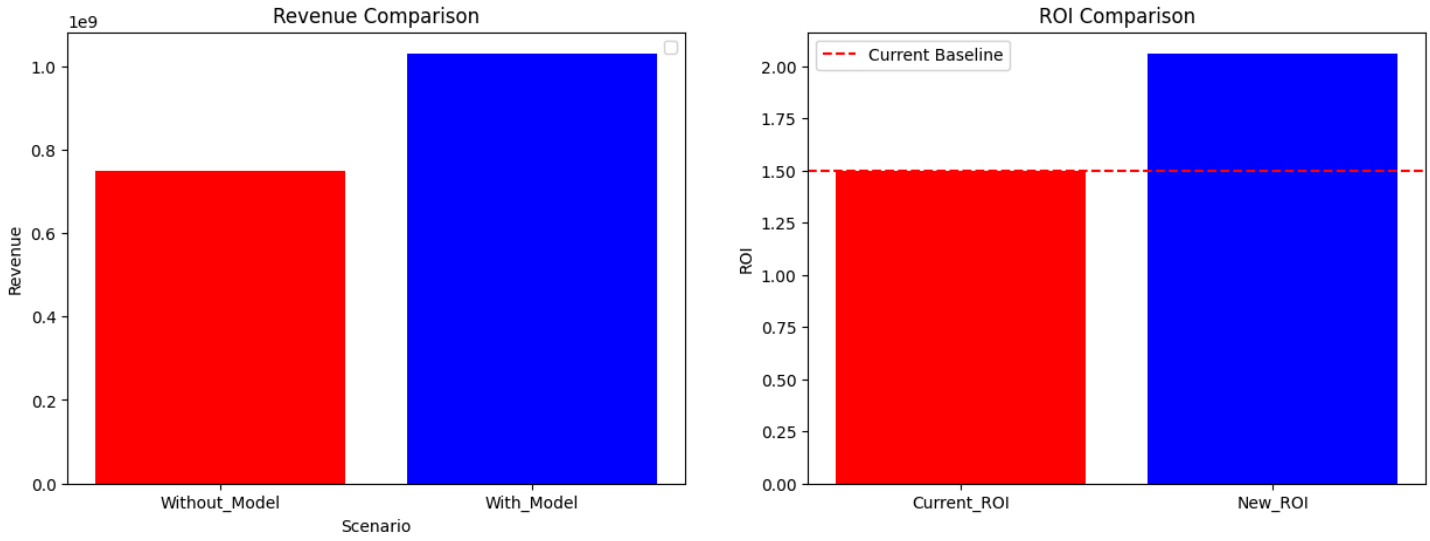

Fig 5. The two visuals above compare the potential change in revenue and ROI if our model was deployed. As you can see, deployment of the prediction model is far more effective than the traditional promotion/marketing allocation strategy.

Appendix and Contributions:	
Comparison Models (Kaggle)	https://www.kaggle.com/code/yasserh/song-popularity-prediction-best-ml-models
Sam MacArthur	Contribution: data-set research, preliminary analysis, model development/building, model testing, business application, report/ppt composition.
Neha Maqsood	Contribution: introduction, preliminary analysis, performance comparison with benchmark, ppt
Krishna Gupta	Contribution: data-set research, preliminary analysis , report/ppt composition
Chris Chen	Contribution: preliminary analysis, business application focus, report/ppt composition
Ziling Chen	Contribution: data-set research, feature analysis, report/ppt composition